KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 1

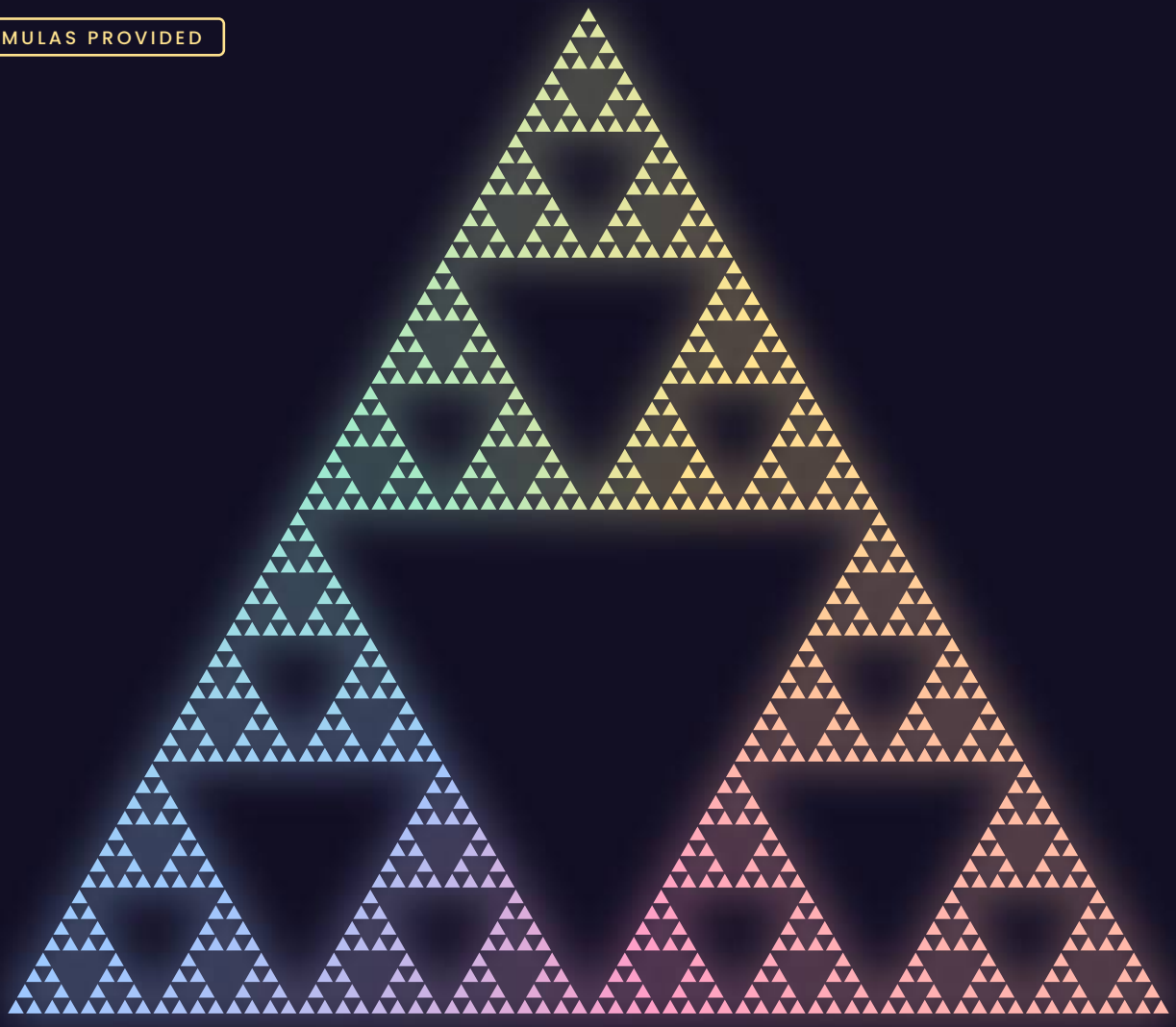
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 10

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

SIERPIŃSKI TRIANGLE
three copies of itself, each half the size

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

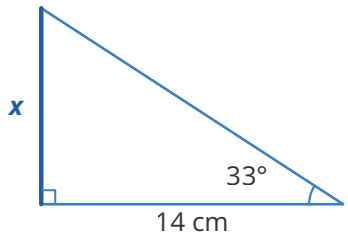
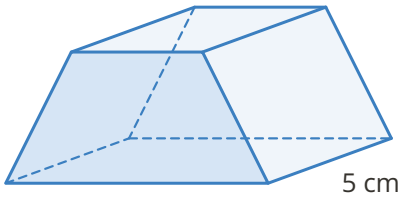
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 The shaded face has an area of 24 cm^2. Find the volume of the prism.  	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER
3 Find Arjun's commission. Value of sales: $120 000 Commission rate: 3%	FORMULA Commission = percentage × value of sales COPY SUBSTITUTE ANSWER
4 Share $90 in the ratio 2 : 7. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is white. Colour: Red, White Marbles: 5, 3	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

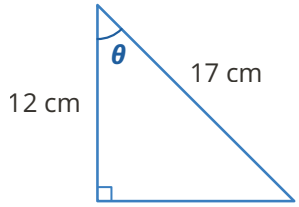
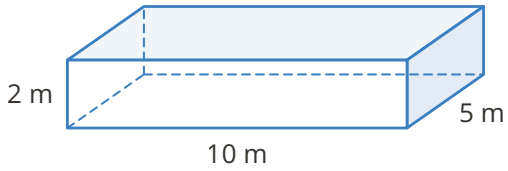
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree. 	FORMULA $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$ COPY SUBSTITUTE ANSWER
2 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER
3 Find the percentage loss. Cost price \$120 Selling price \$90	FORMULA Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$ COPY SUBSTITUTE ANSWER
4 Simplify. $y^9 \div y^4$	FORMULA $a^m \div a^n = a^{m-n}$ COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 5$. $y = 2x - 3$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

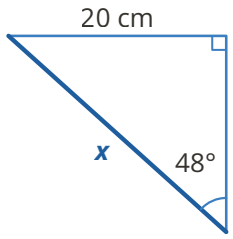
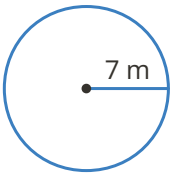
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find the new population of the town. Population 2400 Increase 5%	FORMULA New amount = original × (100% + increase) COPY SUBSTITUTE ANSWER
4 Find h when $A = 24$ and $b = 8$. $A = \frac{1}{2}bh$	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
5 Find the probability that the bus is not late. $P(\text{late}) = 0.15$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 2

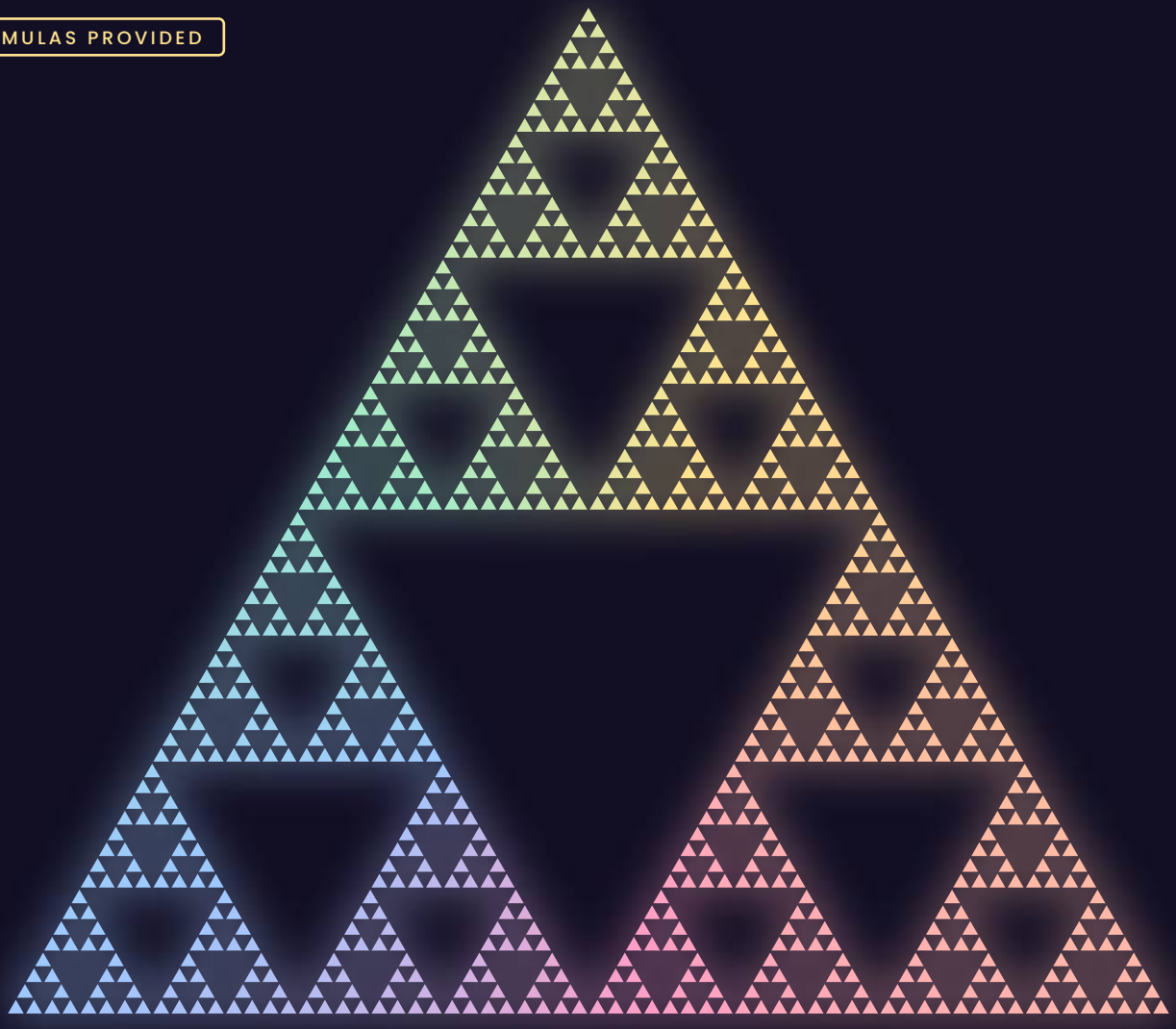
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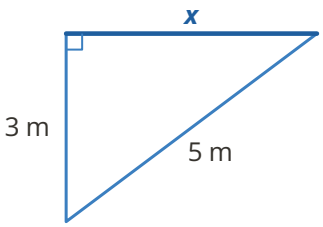
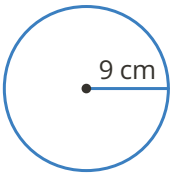
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a^2 = c^2 - b^2$ COPY SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER
3 Find the new weekly wage. Weekly wage \$900 Pay rise 4%	FORMULA New amount = original × (100% + increase) COPY SUBSTITUTE ANSWER
4 Simplify. $12a^5 \div 4a^2$	FORMULA $a^m \div a^n = a^{m-n}$ COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is blue. Colour: Blue 6, Green 4, Red 2	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

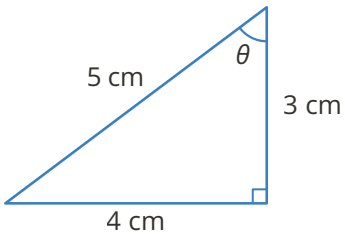
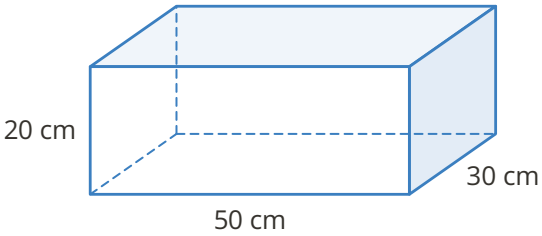
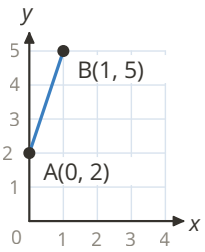
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\cos\theta$ as a fraction.  	FORMULA $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$ COPY SUBSTITUTE ANSWER
2 How many litres does this tank hold?  	FORMULA $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$ COPY SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: \$2000 Interest rate: 4% per year Time: 3 years	FORMULA $I = Prn$ (interest = amount × yearly interest rate × number of years) COPY SUBSTITUTE ANSWER
4 Simplify. $6a^0$	FORMULA $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$ COPY SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

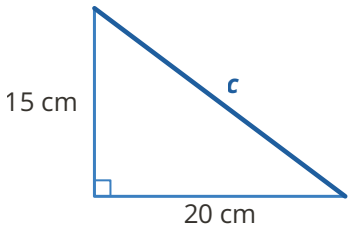
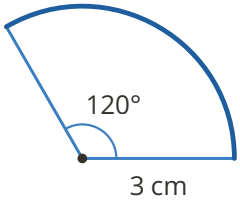
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 2 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA $l = \frac{\theta}{360} \times 2\pi r$ COPY SUBSTITUTE ANSWER
3 Find the original price of the shirt. Discount: 20% Sale price: $48	FORMULA Original amount = amount ÷ its percentage × 100 COPY SUBSTITUTE ANSWER
4 Write in scientific notation. 7000	FORMULA $a \times 10^n$, where $1 \leq a < 10$ COPY SUBSTITUTE ANSWER
5 Find the value of y when $x = 2$. $y = 6x + 1$	FORMULA $y = mx + c$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 3

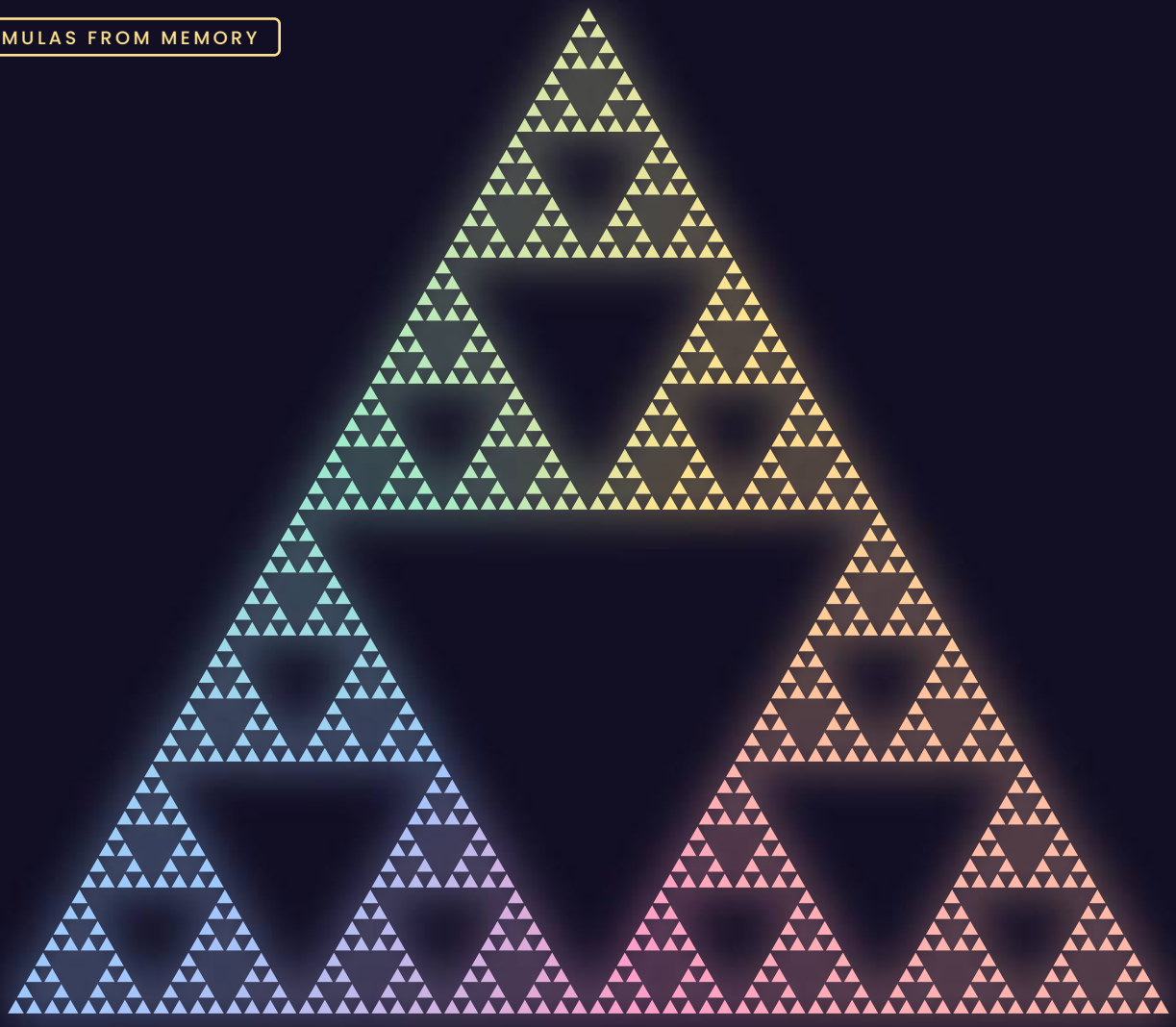
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ACHIEVEMENT THROUGH ENDEAVOUR

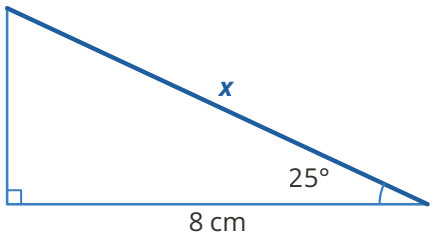
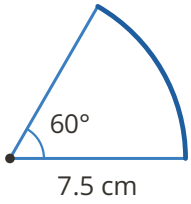
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Ruby earns a salary of \$91 000 a year. Find Ruby's weekly pay.	FORMULA SUBSTITUTE ANSWER
4 Write as a single power. $(5^2)^3$	FORMULA SUBSTITUTE ANSWER
5 Find the value of y when $x = -4$. $y = -x + 6$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

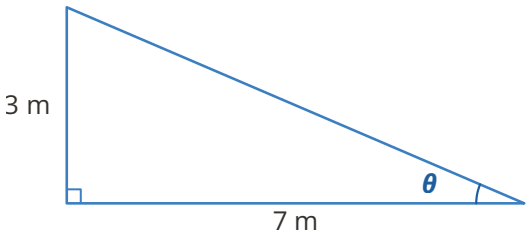
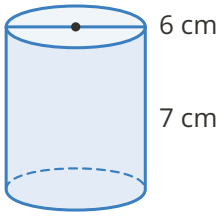
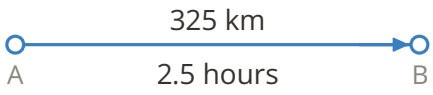
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the percentage loss. Cost price \$125 Selling price \$115	FORMULA SUBSTITUTE ANSWER
4 A train travels from A to B. Find the average speed.  	FORMULA SUBSTITUTE ANSWER
5 Find the mean of these scores, correct to 1 decimal place. Score: 2, 3, 4, 5, 6 Frequency: 2, 5, 9, 6, 3	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

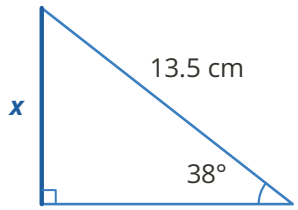
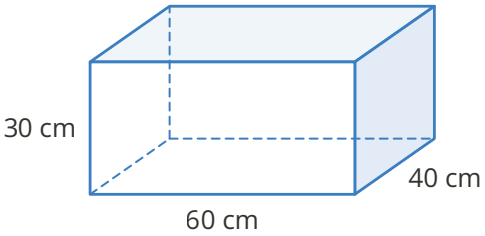
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 3 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 How many litres does this tank hold? 	FORMULA SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: $3500 Interest rate: 2.8% per year Time: 4½ years	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 0.9	FORMULA SUBSTITUTE ANSWER
5 Find the probability that the team does not win. $P(\text{win}) = 0.35$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 4

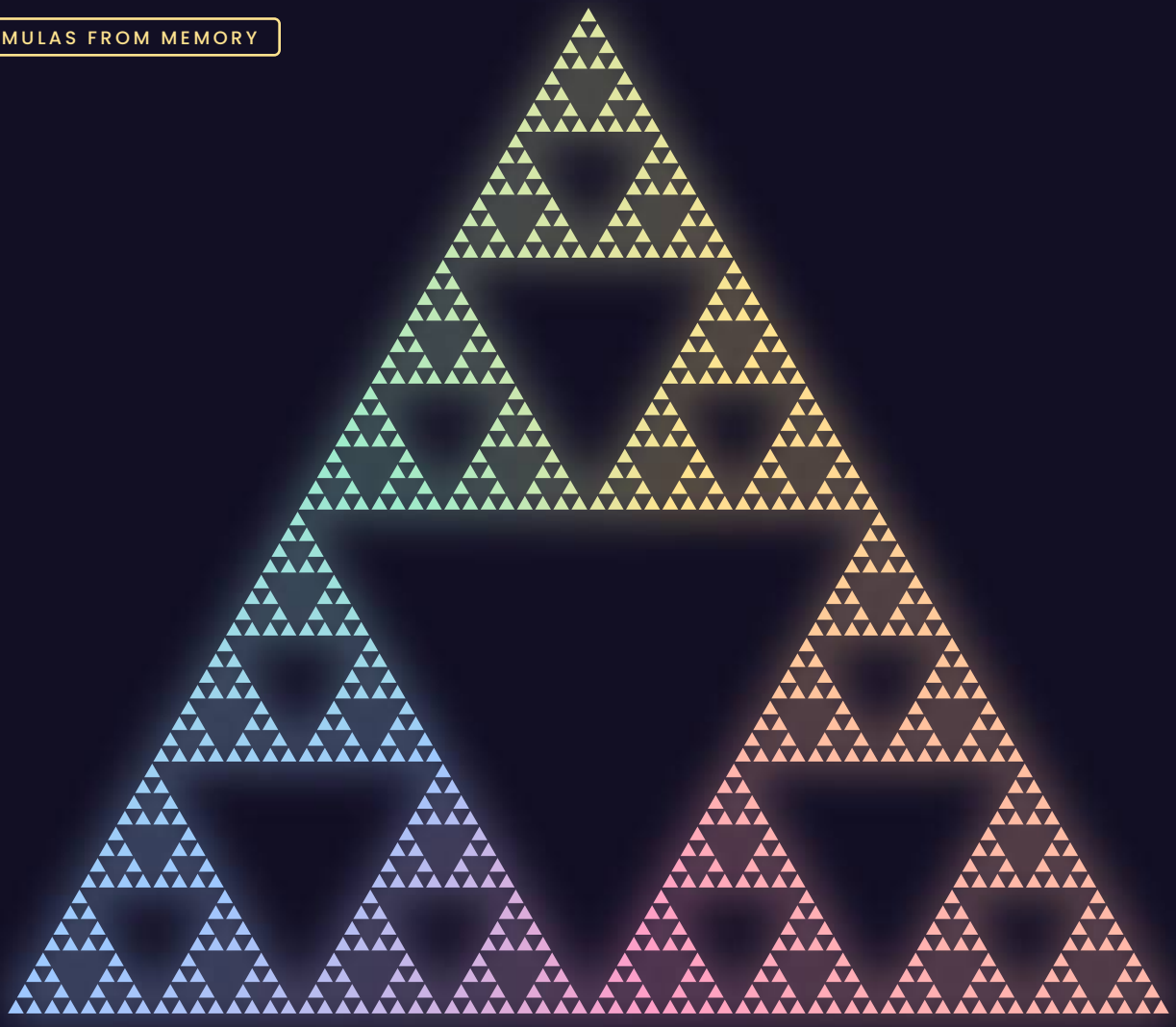
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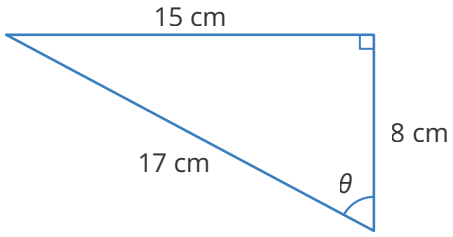
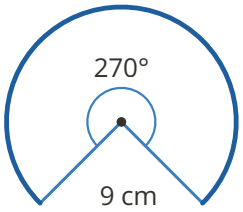
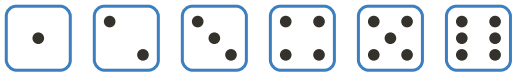
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\sin\theta$ as a fraction.  	FORMULA SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Mia earns \$30 an hour and works 15 hours. How much does Mia earn?	FORMULA SUBSTITUTE ANSWER
4 Find v when $u = 5$, $a = 3$ and $t = 4$. $v = u + at$	FORMULA SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling an even number. 	FORMULA SUBSTITUTE ANSWER

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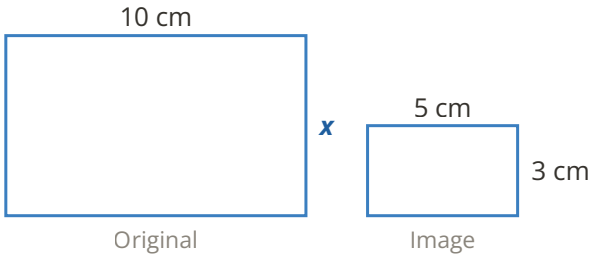
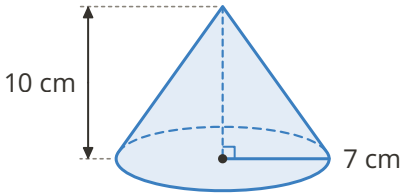
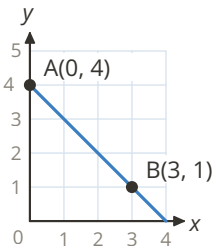
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cone, correct to 1 decimal place.  	FORMULA $V = \frac{1}{3}\pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find the percentage profit. Cost price: $30 Selling price: $45	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 820	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA SUBSTITUTE ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

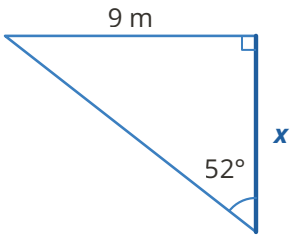
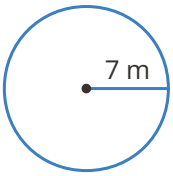
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Ella earns a salary of $68 500 a year. Find Ella's weekly pay, to the nearest cent.	FORMULA SUBSTITUTE ANSWER
4 Write with a positive index. $5k^{-3}$	FORMULA SUBSTITUTE ANSWER
5 Find the mean of these scores, correct to 1 decimal place. Score: 10, 11, 12, 13, 14 Frequency: 2, 6, 4, 5, 1	FORMULA SUBSTITUTE ANSWER